



DEPARTMENT OF ELECTRICAL AND INFORMATION ENGINEERING

UNIVERSITY OF RUHUNA

Module Information Sheet

Code	EC5207	Title	DevOps Engineering		
Credits	2	Total Lectures (Hr)	21	Core/TE/GE	TE
GPA/NGPA	GPA	Total Lab/Workshop/Field Visit/Assignments (Hr)	14	Pre-requisites	None
Aim		The aim is to enable students to use cutting edge software development (Dev) and IT operations (Ops) technologies to automate application deployment on scalable infrastructure.			
Learning Outcomes		Upon successful completion of this module, students will be able to: LO-1 Describe DevOps practices and its function in modern software development organisations. LO-2 Describe Networking and Cloud deployment fundamentals for DevOps Engineering. LO-3 Apply Scripting, Linux and Virtualization knowledge for creating DevOps infrastructure. LO-4 Apply DevOps Concepts such as version controlling, IaC, Infrastructure Automation, CI/CD, Monitoring for build automation. LO-5 Use containerization and orchestration for deploying applications.			
Outline		Infrastructure as Code: Different Languages for Infrastructure as a Code. Developing IaC with Terraform. Provision AWS infrastructure using Terraform. Infrastructure Automation: Remote Configuration Management Concepts. Using Ansible for Configuration Management. CI/CD Pipelines: CI/CD Concept and Tools set. Jenkins Pipelines for CI/CD. Using groovy language for writing Jenkins pipelines.			

	Containerization and Orchestration: Comparison Virtualization/Containerization and Orchestration Dockerizing the applications with different stacks. Microservices and deploying applications on Kubernetes. Application Monitoring: Collecting Different Metrics for Application Monitoring. Application Monitoring Tools set for DevOps.
Methods of Assessments and Marks Allocation	● Continuous Assessments 50% a) Project design document [30%] b) 2 x Project Progress Evaluations [20%] ● End Semester Examination 50% a) Final Project Evaluation [50%]
Eligibility to Appear for the End Semester Examination	● Minimum of 80% attendance for theory classes ● Completion of all components of Continuous Assessments
Necessary Conditions to Pass the Module	● Minimum 35% marks in Continuous Assessments ● Minimum 35% marks in End-Semester Evaluation ● Minimum 40% marks in overall marks
Recommended Readings	● M. Krief, (2022). <i>Learning DevOps A Comprehensive Guide to Accelerating DevOps Culture Adoption with Terraform, Azure DevOps, Kubernetes, and Jenkins</i>. Birmingham: Packt Publishing, Limited. ● G. Kim, P. Debois, J. Willis, J. Humble, and J. Allspaw, (2017). <i>The DevOps handbook : how to create world-class agility, reliability, and security in technology organizations</i>. Portland, Or: It Revolution Press, Llc.